High-Speed Thermosonic Ball Bonder

Authoritative Model Technical Manual & Diagnostic Reference | VAI-MAN-WB-001

NOTICE: SYNTHETIC PROTOTYPE TECHNICAL MANUAL

This document is artificially generated for the VectorAI demonstration and software development.

The specifications, thresholds, service-life values, maintenance intervals, and operating parameters are synthetic and must not be used for real industrial equipment operation or maintenance.

1. Machine Overview & Manufacturing Context

Equipment Model:	High-Speed Thermosonic Ball Bonder	**Document ID:**	VAI-MAN-WB-001
Machine Type Key:	wire-bonding	**Cleanroom Bay / Stage:**	Bay 3B: Wire Bonding Cleanroom
Document Version:	1.0	**Release Date:**	2026-08-27

Ultra-high-speed thermosonic ball bonder designed for creating interconnecting micro-wires (18µm - 32µm gold and copper) between die bond pads and leadframe fingers at speeds up to 24 wires/second.

Process Flow: An Electronic Flame-Off (EFO) spark creates a Free Air Ball (FAB) at the tip of the ceramic capillary. The ultrasonic horn clamps the ball onto the die pad under controlled force and 138 kHz ultrasonic acoustic vibration. The capillary loops over to the leadframe stitch, forms the tail bond, and severs the wire.

2. Major Subsystems & Critical Components

Subsystem / Component	Primary Function	Key Parameters	Degradation Indicators
Ultrasonic Transducer Stack	Converts 138 kHz electrical drive into mechanical acoustic shearing energy.	Acoustic resonance frequency (kHz), vibration amplitude (mm/s), PZT temperature (°C).	Resonance frequency shift (>2.0 kHz), vibration spike (>2.0 mm/s), thermal runaway.
Ceramic Capillary Tool	Guides wire, shapes initial ball bond, and stitches second bond.	Hole diameter (µm), tip chamfer angle, capillary wear index (µm).	Capillary tip erosion, wire drag, golf-ball FAB shape.
EFO Spark Electrode	Discharges high-voltage arc to melt wire tail into uniform spherical ball.	Spark voltage (V), spark current (mA), arc gap distance (mm).	EFO firing timeouts, oxidized Free Air Ball (FAB).
Workholder Heater Block	Maintains leadframe temperature at 150-220°C for intermetallic bonding.	Platen temperature (°C), leadframe clamping force (N).	Thermal fluctuation, leadframe micro-bouncing during ultrasonic burst.

3. Sensor Telemetry & Threshold Matrix

Threshold boundaries configured for real-time telemetry monitoring and automatic anomaly triggers:

Sensor Name & ID	Unit	Normal Range	Warning Range	Critical Range	Direction
Ultrasonic Vibration vibration_ultrasonic	mm/s	0.35 – 0.65	**0.65 – 1.8**	**1.8 – 5**	Higher is Worse
Transducer Head Temp temperature_transducer	°C	40 – 52	**52 – 65**	**65 – 100**	Higher is Worse

4. Deterministic RUL Model (Weighted Degradation)

Base Useful Life: 1,650 operating hours. **Model:** RUL = BaseLife * (1 - (0.55*wear_vib + 0.30*wear_temp + 0.15*wear_clamp))

Parameter	Sensor ID	Unit	Weight	Healthy Limit	Critical Limit	Direction
Ultrasonic Vibration	vibration_ultrasonic	mm/s	**0.55**	0.5	2.5	HIGHER IS WORSE
Transducer Temp	temperature_transducer	°C	**0.30**	48	68	HIGHER IS WORSE
Clamp Force	load_clamp	N	**0.15**	68	90	HIGHER IS WORSE
TOTAL WEIGHT SUM	-	-	**1.00 (100%)**	-	-	VERIFIED OK

5. Troubleshooting & Anomaly Symptoms Matrix

Symptom & ID	Severity	Related Sensors	Possible Root Causes	Recommended Corrective Action
Non-Stick On Pad (NSOP) Defect Rate > 0.05% SYM-WB-001	**CRITICAL**	vibration_ultrasonic, temperature_transducer	• Capillary tip erosion • PZT resonance frequency shift • Bond pad oxidation	Perform emergency capillary tool replacement and execute 138.4 kHz frequency auto-tune.

6. Authoritative Diagnostic Knowledge Base (Failure Scenarios)

Scenario 1: PZT Transducer Thermal Runaway & Acoustic Resonance Decoupling (SCEN-WB-001)	Severity: CRITICAL
Telemetry Sensor Pattern:	vibration_ultrasonic > 2.5 mm/s, temperature_transducer > 65°C
Possible Root Causes:	Piezoelectric crystal thermal stress | Capillary loose in horn clamp
Recommended Corrective Action:	Immediate tool stop: Replace 25µm capillary tool and recalibrate transducer stack at 138.4 kHz.
Verification & Recovery Steps:	Verify transducer resonance impedance curve > Execute wire pull test (> 6.5g) and ball shear test (> 25g)